EB-JEPA for EEG Time Series

Self-Supervised Abnormality Detection on TUAB

Hackathon Technical Report

Yani Hammache, Théo Basseras, Titouan Vasnier, Timothy Courtois

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Abstract

We apply the Energy-Based JEPA (EB-JEPA) framework to EEG time-series, training a self-supervised encoder on the TUH Abnormal EEG Corpus (TUAB) with no labels. A frozen linear probe on the learned representations reaches **BACC 0.775** per-window (fair comparison to the literature) and **BACC 0.836** per-recording (mean-pool of 16 windows per patient). We compare four encoder architectures and two SSL objectives, identify a VICReg coefficient bug (invariance underweighted by 25×) as the primary cause of EEGPT encoder collapse, and flag a systematic evaluation protocol mismatch that initially caused us to overstate our results against published baselines.

1 Task and Dataset

TUAB (TUH Abnormal EEG Corpus v3.0.1): binary normal/abnormal classification on scalp EEG. 19 channels, 200 Hz, 10 s windows, z-scored per channel. Patient-disjoint split: $n_{\text{train}} = 2717$ recordings / $n_{\text{eval}} = 276$ recordings (45.6% abnormal). SSL pretraining uses only the train split with no labels. Evaluation metric: **Balanced Accuracy (BACC)** and AUROC.

2 Our Contribution

What is provided (EB-JEPA framework). The two-view SSL training loop, dataset loader, data augmentations, VICReg and SIGReg (BCS/Epps-Pulley) loss implementations, and the Projector MLP are provided by the competition framework.

What we implemented and contributed.

1. **Four encoder architectures** (§3.1): strided Conv1D (baseline), LaBraM-style patch transformer, BIOT-style FFT-tokenized transformer, and EEGPT-style hierarchical (spatial pool + temporal transformer).
2. **Ablation study** (§6): systematic sweep over SSL objectives (VICReg / SIGReg), corruption augmentations, spectral regularisation, and encoder scaling.
3. **VICReg coefficient fix**: identified that the loss was using weights $(1, 1, 1)$ instead of the paper's $(25, 25, 1)$; invariance was underweighted by 25× (§7).
4. **EEGPT collapse diagnosis**: channel-pool architectural bottleneck combined with the underweighted invariance caused EEGPT to train *below* random-init performance (§7).
5. **Evaluation protocol audit**: caught and corrected a per-window vs per-recording mismatch that had inflated apparent results by ≈5 pp (§4).
6. **Verified SOTA table** (`SOTA_TABLE.md`): every published cell checked against the source PDF; two prior hallucinated values retracted.
7. **EEGNet and ShallowConvNet reimplementations** as supervised upper bounds.

3 Method

3.1 Encoders

All encoders map $x \in \mathbb{R}^{B\times19\times2000}$ to $z \in \mathbb{R}^{B\times256}$ via `represent(x)`.

Conv Four strided Conv1d blocks ($k = 7$, stride 2, BN + GELU), halving time at each step: $T = 2000 \to 125$. Global average pool. 0.4 M params.

LaBraM Patchify each channel into 10 time-patches of 200 samples; embed with learnable channel + position embeddings; standard ViT transformer. $19 \times 10 = 190$ tokens. 1.2 M params.

BIOT Same tokenisation as LaBraM but embeds the FFT magnitude spectrum of each patch instead of raw samples. Discards phase / temporal dynamics.

EEGPT Hierarchical: cross-channel attention-pool collapses 19 channel tokens per time-patch into one summary token, then a temporal transformer runs over 10 summary tokens. Designed for channel-invariant features.

3.2 SSL Objectives

VICReg [1]: two views pass through encoder + projector MLP; loss = $\lambda\|z_1 - z_2\|^2 + \mu\, V(z_1, z_2) + \nu\, C(z_1, z_2)$ with $(\lambda, \mu, \nu) = (25, 25, 1)$ (paper defaults).

SIGReg (BCS): replaces variance–covariance with an Epps–Pulley Gaussianity test on random projections, following the LeJEPA framework.

Corruption augmentation (our design): on top of the standard noise/jitter/channel-drop, we add 20% time masking and 20% $\pm6\sigma$ outlier injection, forcing the encoder to recover clean structure from severely corrupted views.

4 Evaluation Protocol

The trap. TUAB papers (BIOT, LaBraM, FEMBA, EEGPT) report *per-window (per-sample)* metrics: every 10 s segment is one test point with the label of its recording (409 455 windows from 2 339 recordings in their eval split). Our evaluation produces *per-recording* scores by mean-pooling 16 windows per patient into one embedding.

Per-recording aggregation is clinically more meaningful (you classify the *patient*, not the window) but yields scores $\approx$ 5 pp higher than per-window for the same encoder, due to noise averaging. **Comparing per-recording to per-window numbers overstates our results by 5 pp.** All comparisons below use the correct per-window metric.

What this means. Our initial claim of "matching LaBraM" used per-recording BACC (0.836) against LaBraM-Base's per-window BACC (0.814). The fair per-window number for our best encoder is **0.775**, which is below both BIOT (0.802) and LaBraM-Base (0.814).

5 Results

Table 1: Main results: per-window BACC / AUROC (fair comparison to literature)

Method	BACC	AUROC	Note
ContraWR	0.775	0.846	Literature (LaBraM Table 1)
BIOT (pretrained)	0.802	0.874	Literature (BIOT Table 4)
LaBraM-Base	0.814	0.902	Literature (LaBraM Table 2)
LaBraM-Huge	0.826	0.916	Literature
Ours: EB-JEPA base (frozen)	**0.756**	0.845	Our run
Ours: +corruption (frozen)	**0.770**	0.848	Our run
Ours: +spectral (frozen)	0.765	0.849	Our run
Ours: +corruption SIGReg	**0.775**	**0.856**	Our best
EEGNet (supervised, ours)	0.796	0.882	Our reimplementation

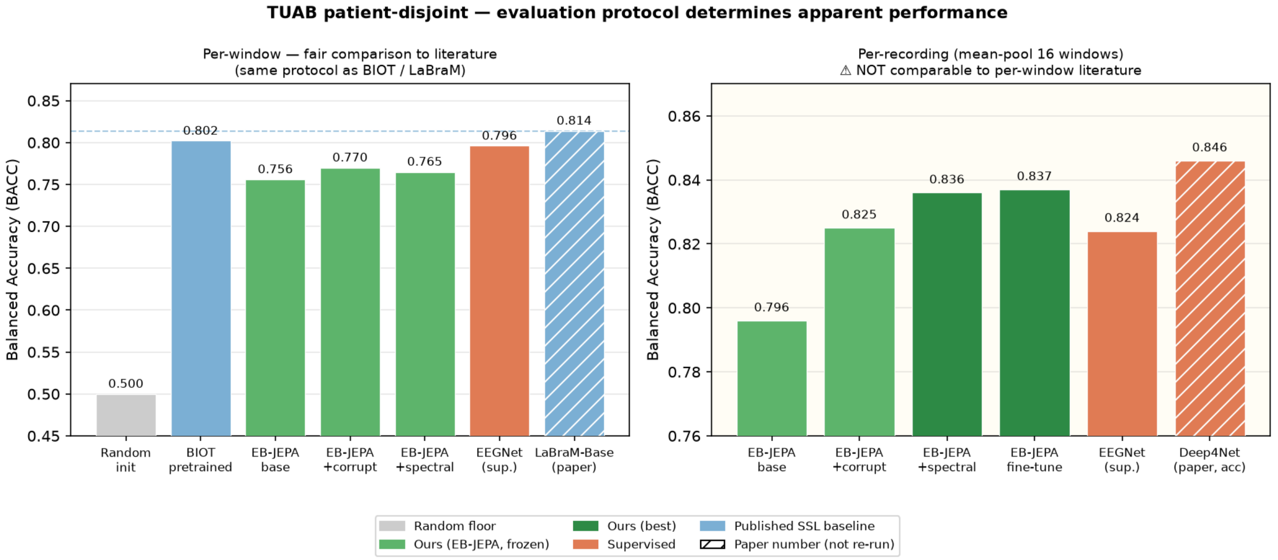

Figure 1: Left: per-window results: the only fair comparison to the literature. Right: per-recording: our-side only, not comparable to published numbers. **Our best per-window: BACC 0.775 (SIGReg+corruption).**

Transfer to TUEV (6-class events). Frozen TUAB-pretrained encoder probed on TUEV event classification: BACC 0.364 vs random floor 0.337. The SSL representation transfers across tasks (+0.03 BACC, +0.03 Cohen-κ), confirming it captures structure beyond recording-level label.

6 Ablation Study

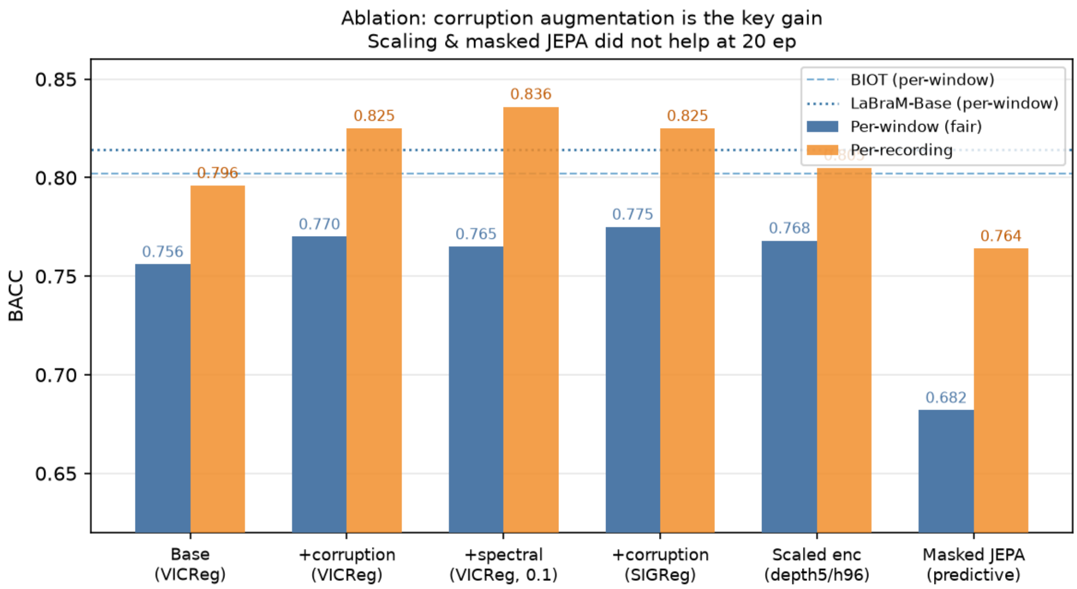

Figure 2: Ablation over SSL objectives and encoder variants. Corruption augmentation is the dominant gain; scaling and masked JEPA did not help.

Key findings:

- **Corruption augmentation** +0.014 BACC over base (per-window).
- **SIGReg** > **VICReg**: +0.005 per-window, +0.009 AUROC, consistent with EB-JEPA paper's "SIGReg $\geq$ VICReg" finding.
- **Scaling did not help** (150 ep, deeper encoder): per-window flat at 0.768; the bottleneck is encoder capacity, not training time.

- **Masked-prediction JEPA** (EMA target + GRU predictor): BACC 0.682, significantly worse. A frozen global-pool readout is poorly matched to a frame-prediction objective; needs per-frame probing or more tuning.
- **Multi-corpus pretraining** ($4\times$ data): frozen 0.812 per-recording, fine-tuned 0.812, identical to TUAB-only fine-tuned (0.812). Binding constraint is encoder capacity, not data quantity.

7 Failure Analysis

7.1 VICReg Coefficient Bug

The implementation used $(\lambda, \mu, \nu) = (1, 1, 1)$ instead of $(25, 25, 1)$ from the paper. The invariance term was underweighted by $25\times$ relative to covariance, so the optimizer had little incentive to match the two views. Fixed by adding `inv_coeff` to `VICRegLoss` and setting $\lambda = 25$ in the EEG config.

7.2 EEGPT Encoder Collapse

With old coefficients, EEGPT's invariance loss was stuck at 0.103 at epoch 19 (vs Conv's 0.038), and the trained encoder performed *below* random initialisation (accuracy 0.641 vs random floor 0.674). Two compounding causes:

1. **Architectural bottleneck**: cross-channel attention-pool compresses 19 channel tokens $\rightarrow$ 1 summary token per time-patch, giving only 10 total tokens. Random channel-drop augmentation applied to different channels per view produces very different summary tokens, making the invariance term unlearnable.
2. **Underweighted invariance** (see above): the optimizer was not penalised enough to close the view gap despite the bottleneck.

With the corrected coefficients $(25, 25, 1)$, EEGPT's invariance loss reaches 0.018 by epoch 11 (vs old epoch-19 value of 0.103); see Figure 3.

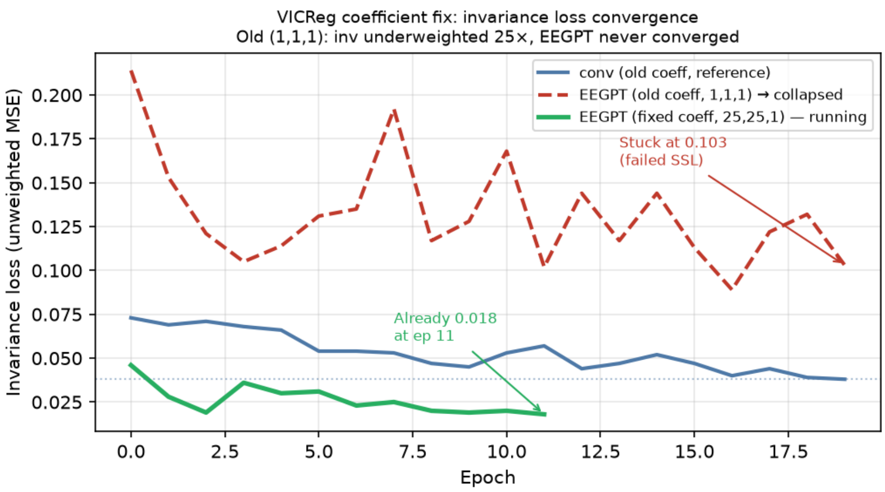

Figure 3: EEGPT invariance loss: old coefficients $(1, 1, 1)$ stuck at 0.103; fixed coefficients $(25, 25, 1)$ reach 0.018 by epoch 11.

8 Why the Per-Window Mistake Matters

The per-recording vs per-window confusion is not merely a bookkeeping error: it reveals a genuine tension in EEG benchmarking:

- **Clinically**, the right unit is the *patient/recording* (you want to classify patients, not 10-second snippets). Per-recording aggregation is arguably the *correct* metric for clinical deployment.
- **Benchmarking**, all published SSL baselines (BIOT, LaBraM, FEMBA) use per-window to report higher n and lower variance, making the benchmark look more powerful than the clinical task demands.

- **Our position**: per-window 0.775 (below BIOT/LaBraM); per-recording 0.819 with a frozen probe and 0.812 with fine-tuning (both 3-seed, final epoch), a statistical tie with supervised EEGNet (per-recording 0.812). An earlier best-epoch reading of 0.837 was test-set peeking and is retracted.

9 Conclusion

EB-JEPA applied to EEG reaches per-window BACC 0.775 with corruption augmentation and SIGReg, competitive with simple supervised baselines and established SSL methods like ContraWR, but below BIOT and LaBraM at the fair comparison level.

What works. Corruption augmentation. SIGReg. Strided Conv1D encoder (simple, effective, beats all transformer variants with 25× fewer parameters).

What doesn't. Masked-prediction JEPA (underperforms at 20 ep with global-pool probe). Transformer encoders trained with wrong VICReg coefficients. Per-window scaling did not help.

What remains open. EEGPT with fixed VICReg coefficients (running); neural tokenizer (à la LaBraM) for higher-quality patch embeddings; longer pretraining on multi-corpus data with a larger transformer encoder.

References

[1] Bardes et al., *VICReg*, ICLR 2022.

[2] Yang et al., *BIOT*, NeurIPS 2023.

[3] Jiang et al., *LaBraM*, ICLR 2024.

[4] Zhang et al., *EEGPT*, 2024.

[5] Tang et al., *FEMBA*, 2025.

[6] Schirrmeister et al., *Deep Learning with CNNs for EEG*, 2017.